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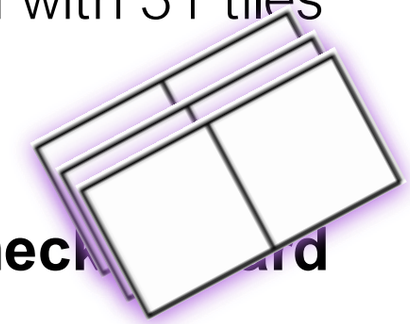
COMP90038 Algorithms and Complexity

Lecture 2: Review of Basic Concepts

(with thanks to Harald Søndergaard)
Olya Ohrimenko / Cezary Kaliszyk

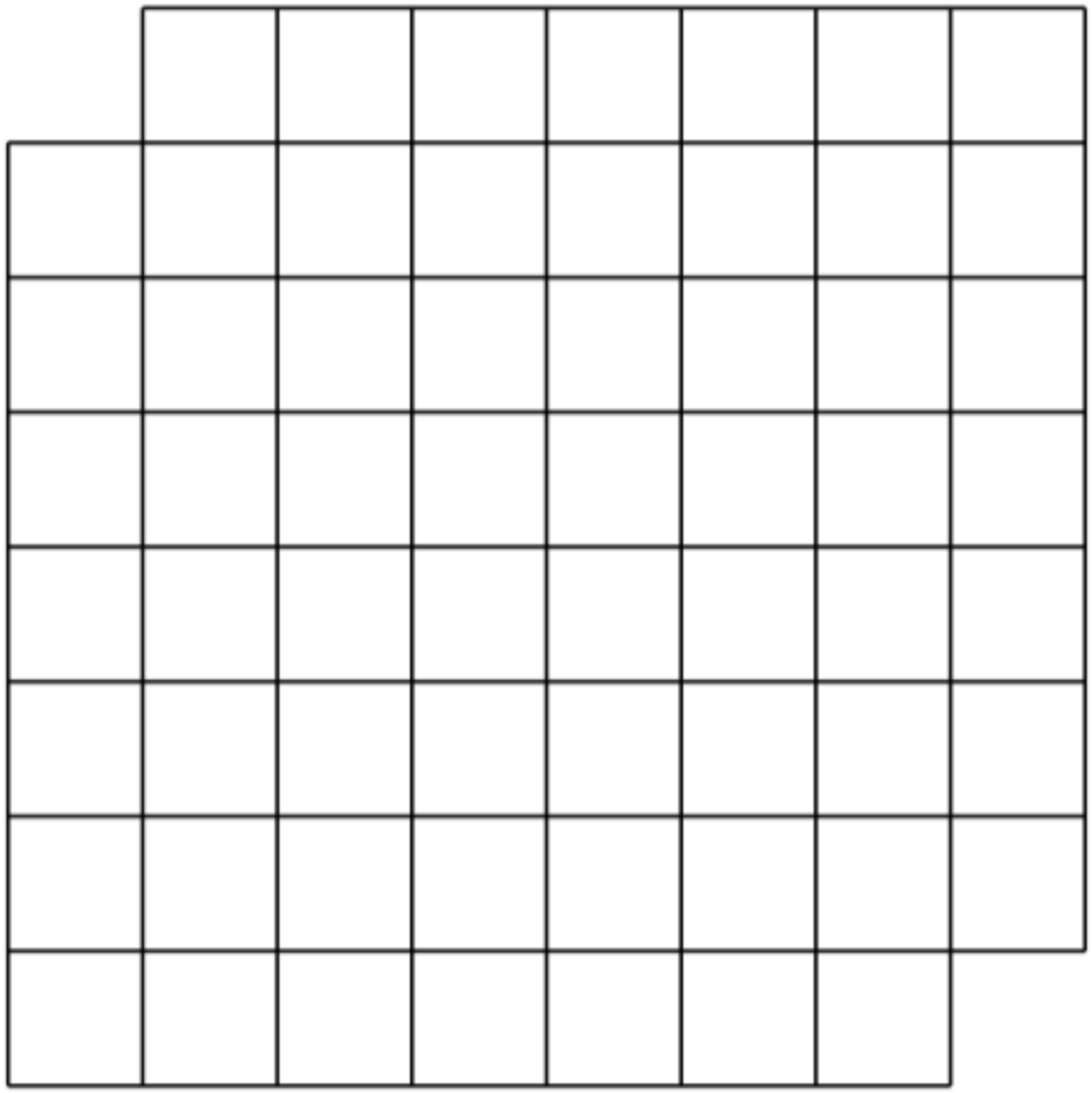
Approaching a problem

- Can we cover this board with 31 tiles of the following form?



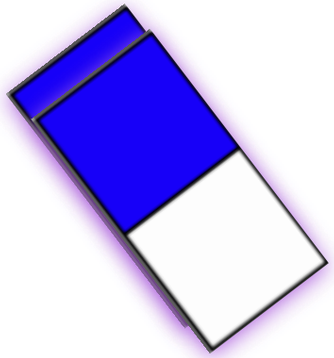
- This is the **mutilated chec rd problem**.

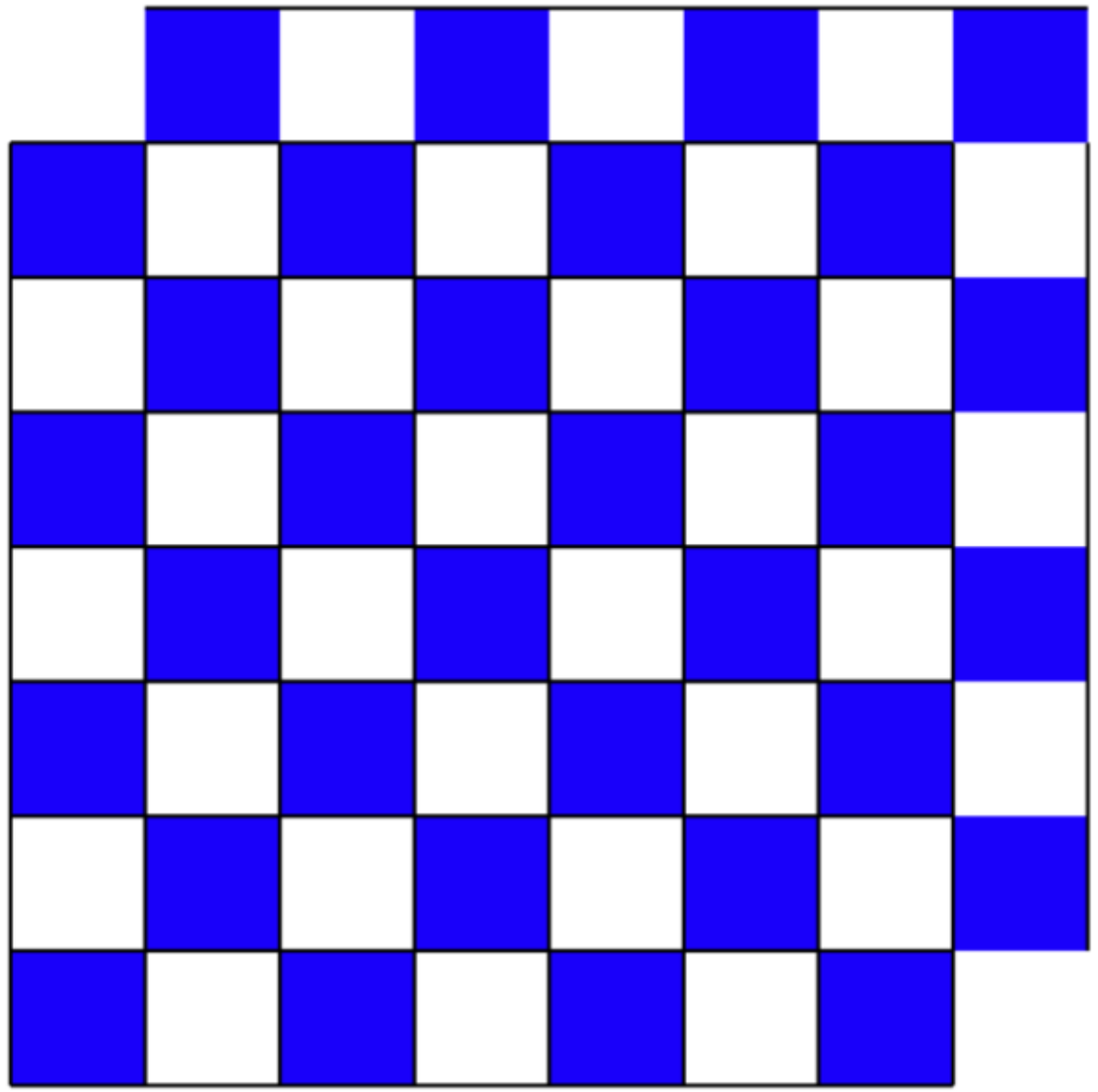
- There are only finitely many ways we can arrange the 31 tiles, so there is a brute-force (and very inefficient) way of solving the problem.



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Transform and Conquer? Use abstraction?





- Can we cover this board with 31 tiles of the form shown?
- Why can we quickly determine that the answer is no?
- **Hint:** Using the way the squares are coloured helps.

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Algorithms and Data Structures

- **Algorithms**: for solving problems, transforming data.

- **Data structures**: for storing data; arranging data in a way that suits an algorithm.

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Algorithms and Data Structures

- **Algorithms**: for solving problems, transforming data.

- **Data structures**: for storing data; arranging data in a way that suits an algorithm.

- Which data structures are you familiar with?

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Algorithms and Data Structures

- **Algorithms**: for solving problems, transforming data.

- **Data structures**: for storing data; arranging data in a way that suits an algorithm.
  - **Linear** data structures: stacks and queues
  - Trees and graphs
  - Dictionaries
- Which data structures are you familiar with?

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Primitive Data Structures: The Array

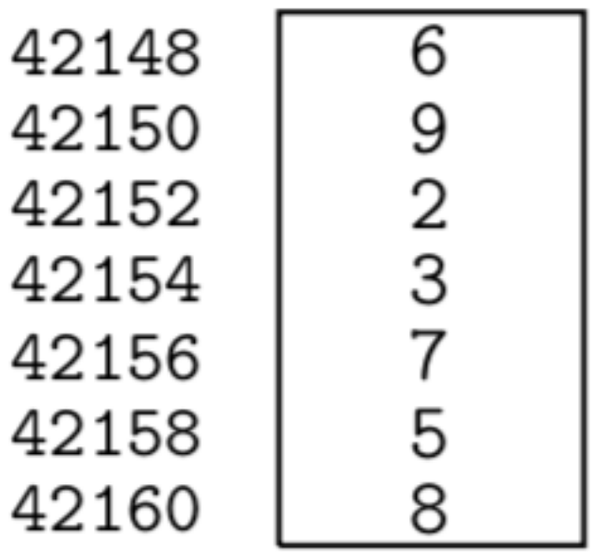
- An array corresponds to a sequence of consecutive cells in memory.
- Depending on programming language: `A[0]` up to `A[n-1]`, or `A[1]` up to `A[n]`.
- Locating a cell, and storing or retrieving data at that cell is very fast.



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The Array in Memory





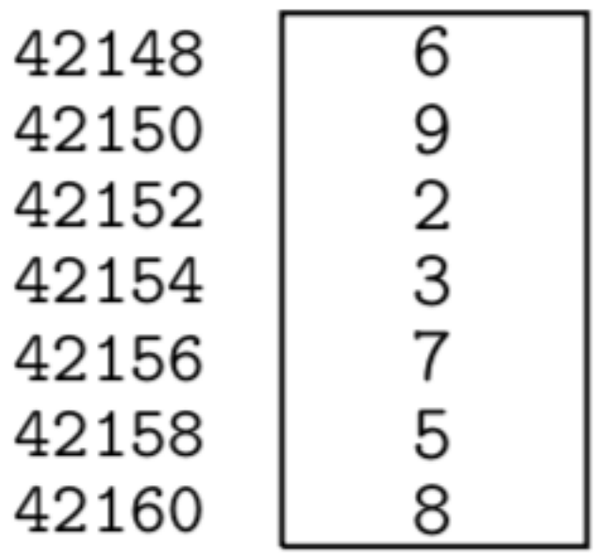
How many bytes does each integer occupy here?

Answer: 2 (16-bit integers)

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Primitive Data Structures: The Array

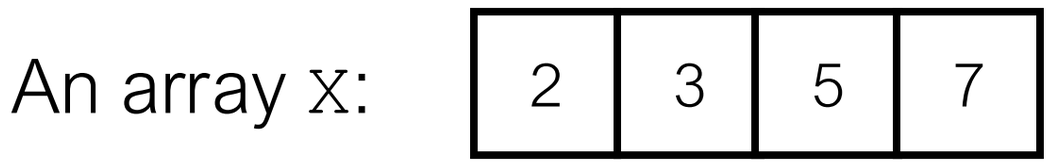


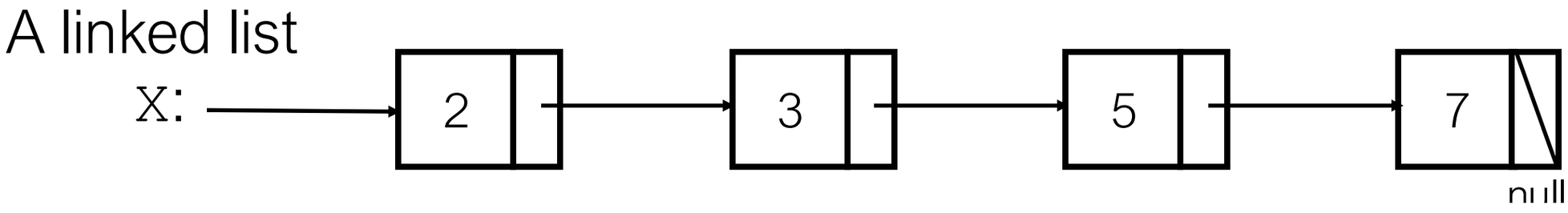


The downside of an array is that maintaining a contiguous bank of cells with information can be difficult and time-consuming.

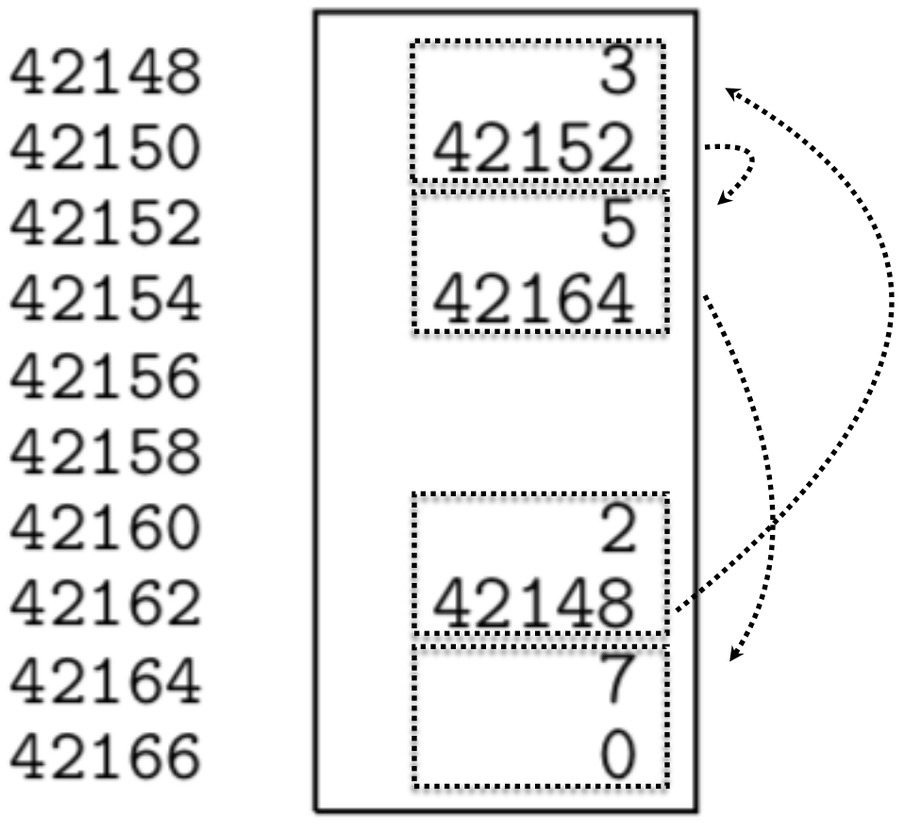
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Primitive Data Structures: The Linked List



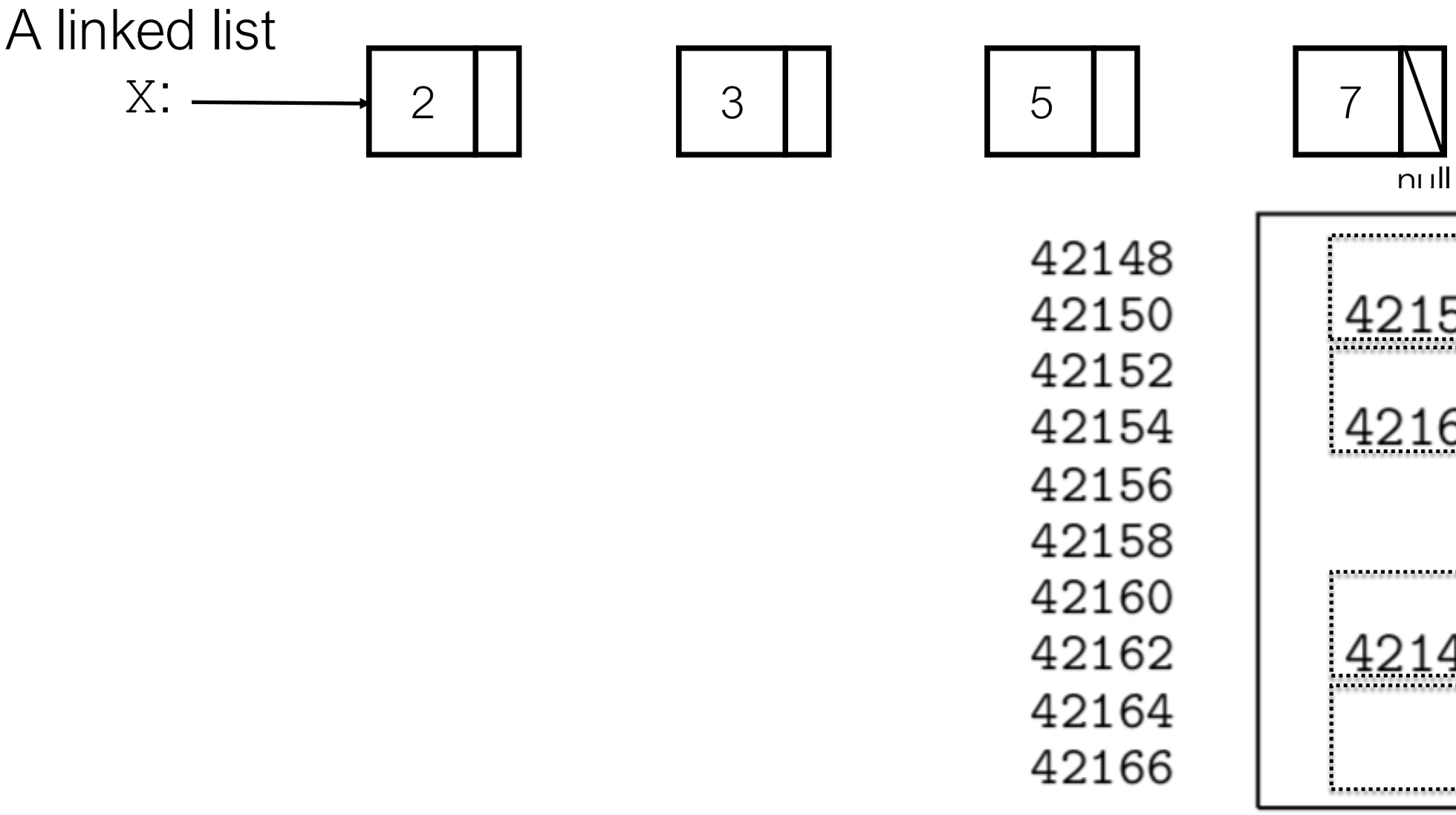


Suppose variable x holds the address 42160, then the list could look like this in memory:



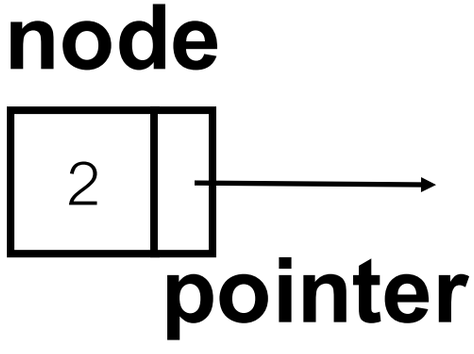
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Primitive Data Structures: The Linked List

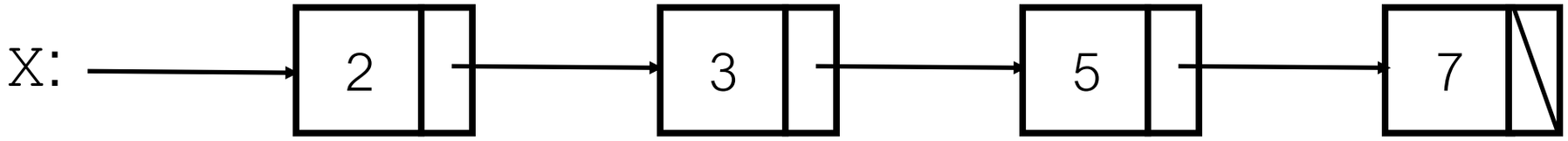


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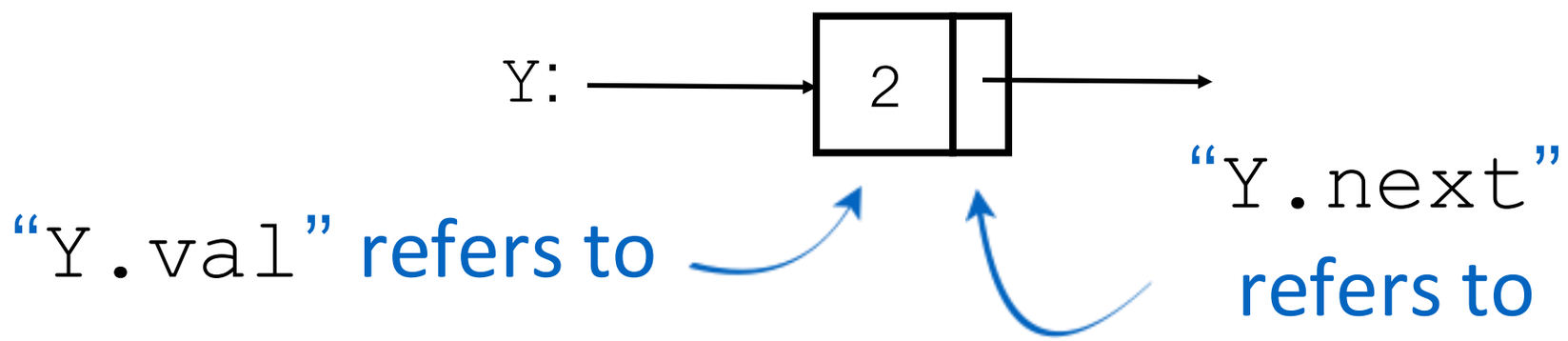
Terminology



(in Java: “reference”)



`X` is (a pointer to) the **head node** of the list



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Linked List

- Often we use a dummy head node that points to the first object, or to a special `null` object that represents an empty list. This makes it easier to write functions that insert or delete elements.

- Inserting and deleting elements is very fast: just move a few links around.

- Finding the *i*th element can be time-consuming.

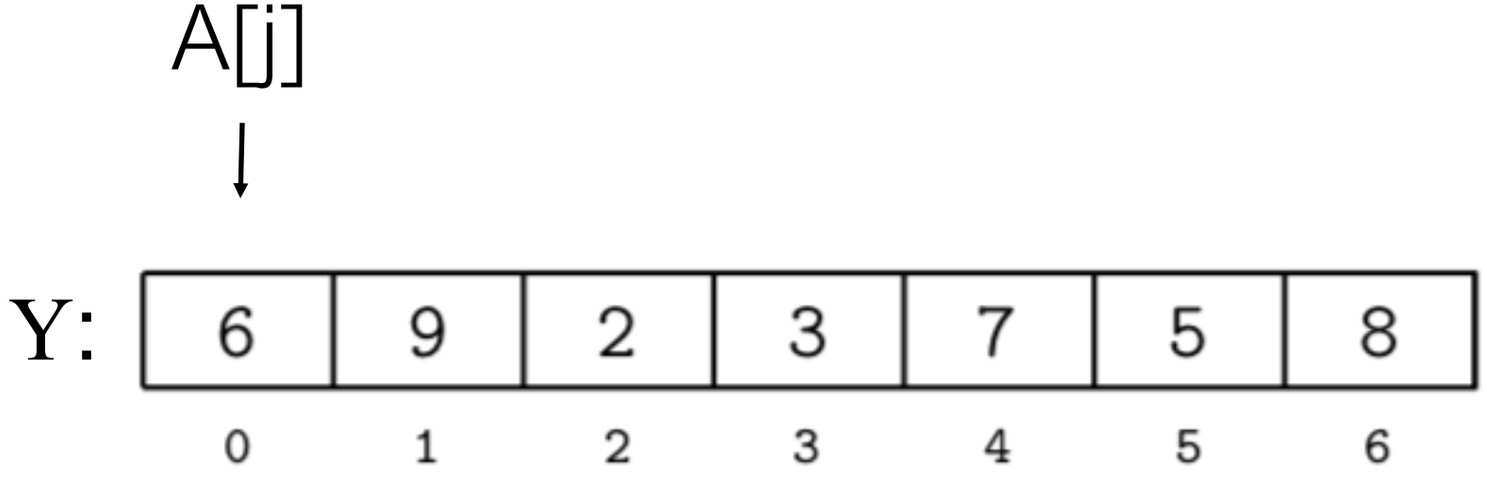
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Iterative Processing: Array

- Walk through the array (of length n)
- For example, to locate item `x`.

```
function find(A,x,n)
    j ← 0
    while j < n
        if A[j] = x
            return j
        j ← j+1
    return -1
```

A: Y x: 7 n: 7 j: 0



Let's trace the execution of find(Y,7,7)

(returns 4)

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Iterative Processing: List

- Walk through a linked list.
- For example, to locate item `x`.

(note similarity to array version)

```
function find(head,x)
    p ← head
    while p ≠ null
      if p.val = x
        return p
      p ← p.next
  return null
```

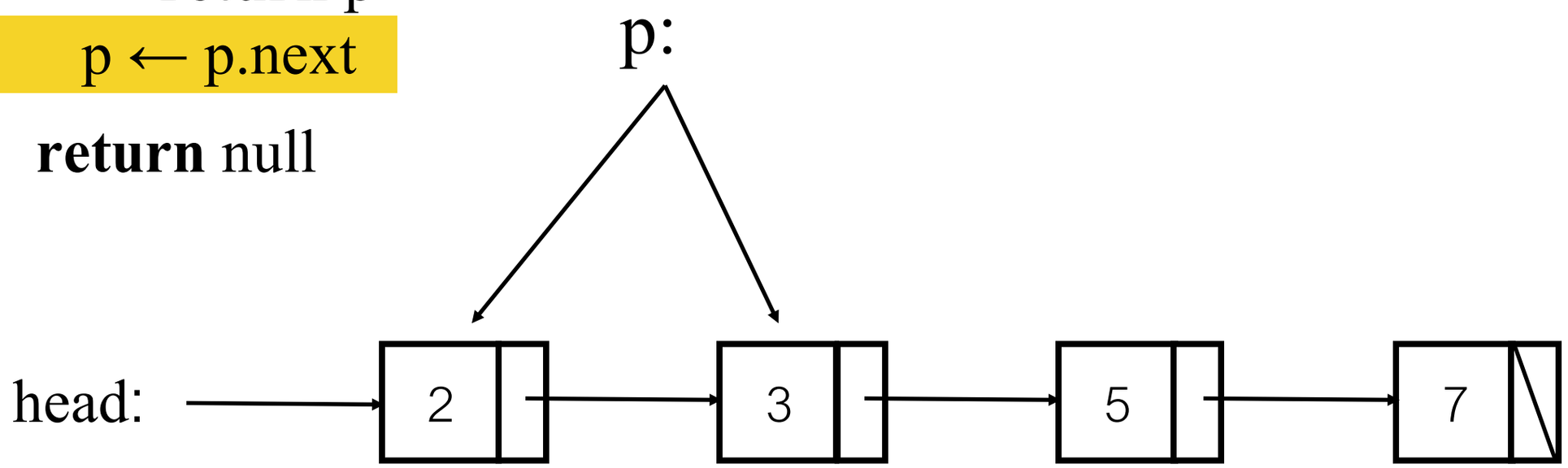
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Iterative Processing: List

- Walk through a linked list.
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```
function find(head,x)
    p ← head
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```



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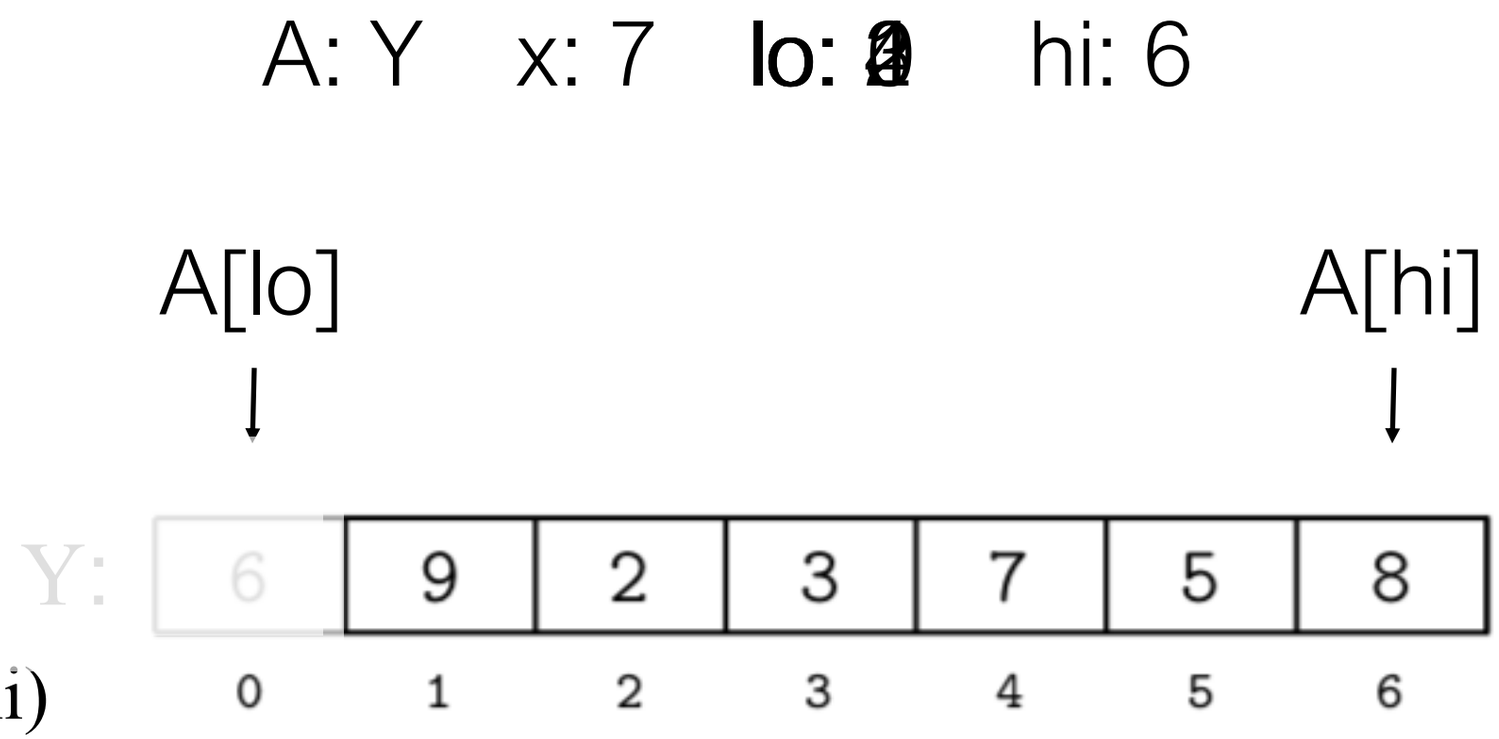
Recursive Processing: Array

Solve the problem for a sub-instance and use the solution to solve the full instance

- For example, to locate item `x`.

```
function find(A,x,lo,hi)
    if lo > hi
        return -1
    else if A[lo] = x
        return lo
    else
        return find(A,x,lo+1,hi)
```

Initial call: find(A,x,0,n-1)



Let's trace the execution of find(Y,7,0,6)

(returns 4)

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Recursive Processing: List

- Solve the problem for a sub-instance and use the solution to solve the full instance

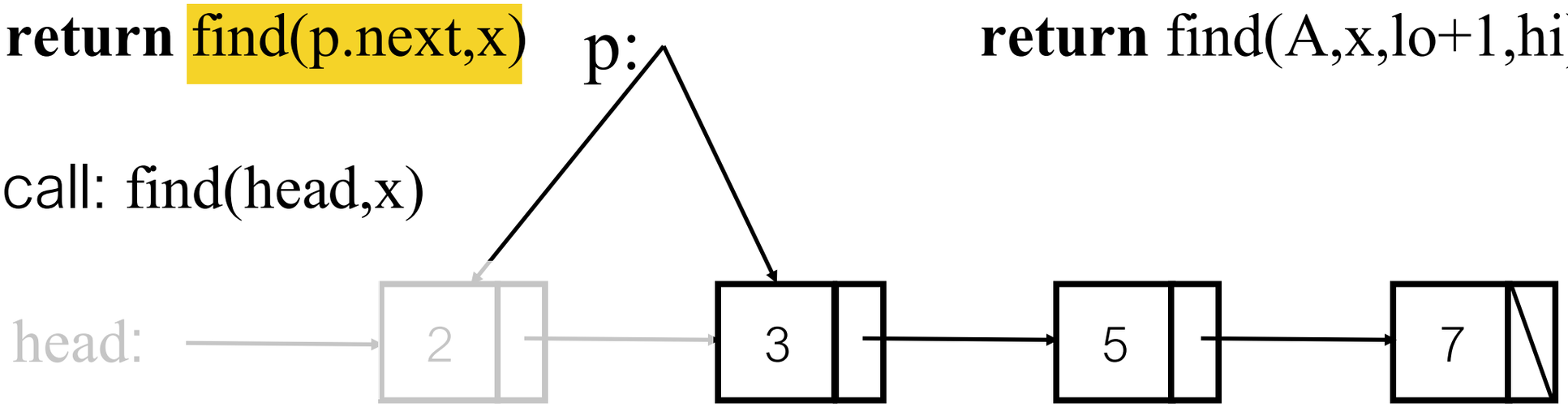
(note similarity to array version)

```
function find(p,x)
    if p = null
        return p
    else if p.val = x
        return p
    else
        return find(p.next,x)
```

We will discuss recursion properly in Week 3

```
function find(A,x,lo,hi)
    if lo > hi
        return -1
    else if A[lo] = x
        return lo
    else
        return find(A,x,lo+1,hi)
```

Initial call: find(head,x)



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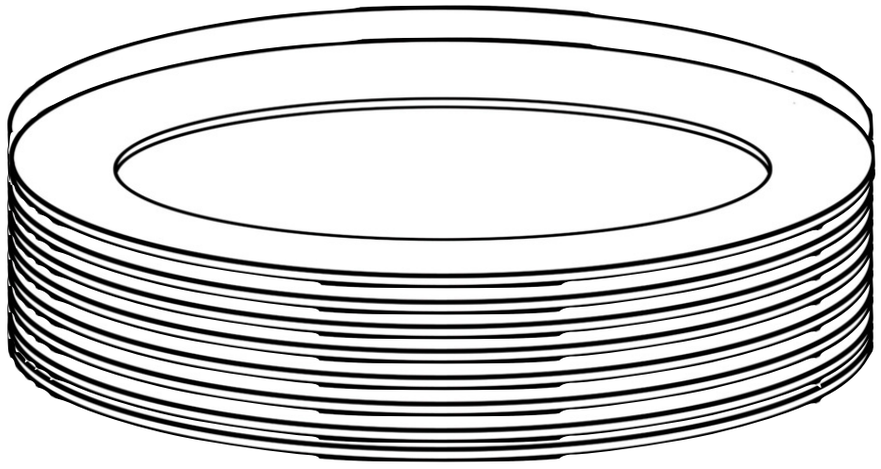
Abstract DataTypes

- A collection of data items, and a family of operations that operate on that data
- Think of an ADT as a set of contracts, an **interface**
- We must still **implement** these promises, but it is an advantage to separate the implementation of the ADT from the “concept” (i.e. the interface it provides)
- Good programming practice is to support this separation
  - Nothing outside of the definition of the ADT should refer to anything inside, except through function calls and basic operations

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Fundamental Data Structure: The Stack

- Last-In-First-Out (LIFO)
- Operations:
  - CreateStack
  - Push
  - Pop
  - Top
  - EmptyStack?
  - …
- Usually implemented as an ADT



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Stack Implementation: Array



top: 4

Push(7)

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Stack Implementation: Array



top: 4

Push(7)

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Stack Implementation: Array

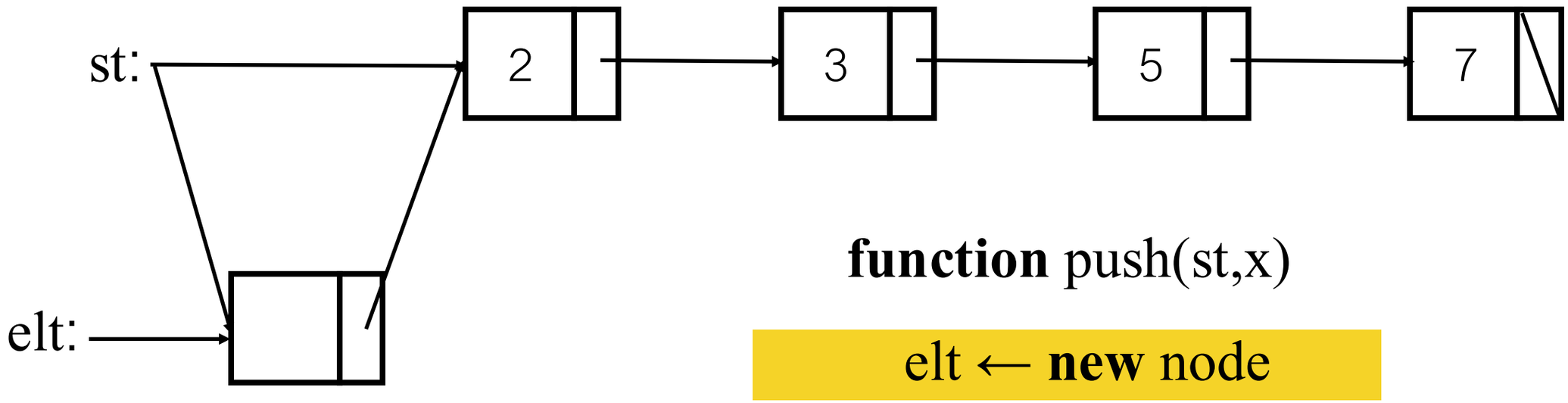


top: 5

Push(7)

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Stack Implementation: Linked List



function push(st,x)

elt ← **new** node

elt.val ← x

elt.next ← st

st ← elt

return st

push(st,6)

See
https://www.cs.usfca.edu/~galles/visualization/Algorithms.html
for more visualisations

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Pseudo Code

- On the previous slide, we assumed that a “node” has two attributes: a “val” which is its value, and a “next” which points to the rest of the list.
- There is no standard for pseudo-code. Use the examples in Levitin as a guide. Cormen et al. pages 20–22 (in Reading Resources) has a list of standard conventions used with pseudo-code which are good to follow, except we use ← as the assignment operator.

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Fundamental Data Structure: Queues

- First-In-First-Out (FIFO)
- Operations:
  - CreateQueue
  - Enqueue
  - Dequeue
  - Head
  - EmptyQueue?
  - …



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Other Data Structures



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- We will meet many other (abstract) data structures, e.g.
  - The priority queue
  - Various types of “tree”
  - Various types of “graph”
- If you check out algorithm animation tools or advanced algorithm books, you will meet exotic data structures such as splay trees and skip lists.

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Summary and Next Week

- Today: Review of Basic concepts (arrays, lists, Abstract Data Types, stack, queue)
- Algorithm analysis—how to reason about an algorithm's resource consumption.

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Announcements

Student representative call:

If you are interested in being a student representative, please sendand email to comp90038-s2-2026@unimelb.edu.au by Wednesday August 5, briefly saying what degree you are doing why you want to be a student rep (one sentence is sufficient).

Don’t forget to do Assumed Knowledge Quizzes

See consultation hours for Assumed Knowledge Quizzes on Canvas

Tutorials start Week 2

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